

PS D:\Hydroponics AI> yolo checks

Ultralytics 8.4.35 Python-3.12.5 torch-2.5.1+cu121 CUDA:0 (NVIDIA GeForce RTX 3050 6GB Laptop GPU, 6144MiB)

Setup complete (12 CPUs, 15.6 GB RAM, 123.2/172.9 GB disk)

OS Windows-11-10.0.26200-SP0

Environment Windows

Python 3.12.5

Install pip

Path C:\Users\HP\AppData\Local\Programs\Python\Python312\Lib\site-packages\ultralytics

RAM 15.65 GB

Disk 123.2/172.9 GB

CPU 13th Gen Intel Core i5-13420H

CPU count 12

GPU NVIDIA GeForce RTX 3050 6GB Laptop GPU, 6144MiB

GPU count 1

CUDA 12.1

numpy 2.4.4>=1.23.0

matplotlib 3.10.8>=3.3.0

opencv-python 4.13.0.92>=4.6.0

pillow 12.1.1>=7.1.2

pyyaml 6.0.3>=5.3.1

requests 2.34.2>=2.23.0

scipy 1.17.1>=1.4.1

torch 2.5.1+cu121>=1.8.0

torch 2.5.1+cu121!=2.4.0,>=1.8.0; sys_platform == "win32"

torchvision 0.20.1+cu121>=0.9.0

psutil 7.2.2>=5.8.0

polars 1.39.3>=0.20.0

ultralytics-thop 2.0.18>=2.0.18

PS D:\Hydroponics AI> yolo predict model=yolo11n-seg.pt

Downloading <https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo11n-seg.pt> to 'yolo11n-seg.pt': 100%  5.9MB 5.5MB/s 1.1s

WARNING 'source' argument is missing. Using default

'source=C:\Users\HP\AppData\Local\Programs\Python\Python312\Lib\site-packages\ultralytics\assets'.

Ultralytics 8.4.35 Python-3.12.5 torch-2.5.1+cu121 CUDA:0 (NVIDIA GeForce RTX 3050 6GB Laptop GPU, 6144MiB)

YOLO11n-seg summary (fused): 113 layers, 2,868,664 parameters, 0 gradients, 9.7 GFLOPs

image 1/2 C:\Users\HP\AppData\Local\Programs\Python\Python312\Lib\site-packages\ultralytics\assets\bus.jpg: 640x480 4 persons, 1 bus, 1 stop sign, 246.9ms

image 2/2 C:\Users\HP\AppData\Local\Programs\Python\Python312\Lib\site-packages\ultralytics\assets\zidane.jpg: 384x640 2 persons, 1 tie, 238.4ms

Speed: 10.8ms preprocess, 242.6ms inference, 56.6ms postprocess per image at shape (1, 3, 384, 640)

Results saved to D:\Hydroponics AI\runs\segment\predict

Learn more at <https://docs.ultralytics.com/modes/predict>

PS D:\Hydroponics AI> dir dataset

Directory: D:\Hydroponics AI\dataset

Mode	LastWriteTime	Length	Name
d-----	26-07-2026 05:38 PM		train

d----- 26-07-2026 05:39 PM valid
-a----- 26-07-2026 05:36 PM 208 data.yaml

PS D:\Hydroponics AI> yolo segment train model=yolo11n-seg.pt data=dataset/data.yaml
epochs=100 imgsz=640 batch=16 device=0 workers=8 project=runs
name=baseline_yolo11n_seg cache=True plots=True patience=20 seed=42

New <https://pypi.org/project/ultralytics/8.4.106> available Update with 'pip install -U ultralytics'

Ultralytics 8.4.35 Python-3.12.5 torch-2.5.1+cu121 CUDA:0 (NVIDIA GeForce RTX 3050 6GB Laptop GPU, 6144MiB)

engine\trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False,
auto_augment=randaugument, batch=16, bgr=0.0, box=7.5, cache=True, cfg=None,
classes=None, close_mosaic=10, cls=0.5, compile=False, conf=None, copy_paste=0.0,
copy_paste_mode=flip, cos_lr=False, cutmix=0.0, data=dataset/data.yaml, degrees=0.0,
deterministic=True, device=0, dfl=1.5, dnn=False, dropout=0.0, dynamic=False,
embed=None, end2end=None, epochs=100, erasing=0.4, exist_ok=False, fliplr=0.5,
flipud=0.0, format=torchscript, fraction=1.0, freeze=None, half=False, hsv_h=0.015,
hsv_s=0.7, hsv_v=0.4, imgsz=640, int8=False, iou=0.7, keras=False, kobj=1.0,
line_width=None, lr0=0.01, lrf=0.01, mask_ratio=4, max_det=300, mixup=0.0, mode=train,
model=yolo11n-seg.pt, momentum=0.937, mosaic=1.0, multi_scale=0.0,
name=baseline_yolo11n_seg, nbs=64, nms=False, opset=None, optimize=False,
optimizer=auto, overlap_mask=True, patience=20, perspective=0.0, plots=True, pose=12.0,
pretrained=True, profile=False, project=runs, rect=False, resume=False, retina_masks=False,
rle=1.0, save=True, save_conf=False, save_crop=False, save_dir=D:\Hydroponics
AI\runs\segment\runs\baseline_yolo11n_seg, save_frames=False, save_json=False,
save_period=-1, save_txt=False, scale=0.5, seed=42, shear=0.0, show=False,
show_boxes=True, show_conf=True, show_labels=True, simplify=True, single_cls=False,
source=None, split=val, stream_buffer=False, task=segment, time=None,
tracker=botsort.yaml, translate=0.1, val=True, verbose=True, vid_stride=1, visualize=False,
warmup_bias_lr=0.1, warmup_epochs=3.0, warmup_momentum=0.8,
weight_decay=0.0005, workers=8, workspace=None

WARNING: All log messages before absl::InitializeLog() is called are written to STDERR

I0000 00:00:1785071193.513485 7364 port.cc:153] oneDNN custom operations are on.
You may see slightly different numerical results due to floating-point round-off errors from
different computation orders. To turn them off, set the environment variable
'TF_ENABLE_ONEDNN_OPTS=0'.

WARNING: All log messages before absl::InitializeLog() is called are written to STDERR

10000 00:00:1785071213.018009 7364 port.cc:153] oneDNN custom operations are on.
You may see slightly different numerical results due to floating-point round-off errors from
different computation orders. To turn them off, set the environment variable
'TF_ENABLE_ONEDNN_OPTS=0'.

TensorBoard: Start with 'tensorboard --logdir D:\Hydroponics
AI\runs\segment\runs\baseline_yolo11n_seg', view at <http://localhost:6006/>

Overriding model.yaml nc=80 with nc=6

	from	n	params	module	arguments
0	-1	1	464	ultralytics.nn.modules.conv.Conv	[3, 16, 3, 2]
1	-1	1	4672	ultralytics.nn.modules.conv.Conv	[16, 32, 3, 2]
2	-1	1	6640	ultralytics.nn.modules.block.C3k2	[32, 64, 1, False, 0.25]
3	-1	1	36992	ultralytics.nn.modules.conv.Conv	[64, 64, 3, 2]
4	-1	1	26080	ultralytics.nn.modules.block.C3k2	[64, 128, 1, False, 0.25]
5	-1	1	147712	ultralytics.nn.modules.conv.Conv	[128, 128, 3, 2]
6	-1	1	87040	ultralytics.nn.modules.block.C3k2	[128, 128, 1, True]
7	-1	1	295424	ultralytics.nn.modules.conv.Conv	[128, 256, 3, 2]
8	-1	1	346112	ultralytics.nn.modules.block.C3k2	[256, 256, 1, True]
9	-1	1	164608	ultralytics.nn.modules.block.SPPF	[256, 256, 5]
10	-1	1	249728	ultralytics.nn.modules.block.C2PSA	[256, 256, 1]
11	-1	1	0	torch.nn.modules.upsampling.Upsample	[None, 2, 'nearest']
12	[-1, 6]	1	0	ultralytics.nn.modules.conv.Concat	[1]
13	-1	1	111296	ultralytics.nn.modules.block.C3k2	[384, 128, 1, False]
14	-1	1	0	torch.nn.modules.upsampling.Upsample	[None, 2, 'nearest']
15	[-1, 4]	1	0	ultralytics.nn.modules.conv.Concat	[1]
16	-1	1	32096	ultralytics.nn.modules.block.C3k2	[256, 64, 1, False]
17	-1	1	36992	ultralytics.nn.modules.conv.Conv	[64, 64, 3, 2]
18	[-1, 13]	1	0	ultralytics.nn.modules.conv.Concat	[1]

19	-1 1	86720	ultralytics.nn.modules.block.C3k2	[192, 128, 1, False]
20	-1 1	147712	ultralytics.nn.modules.conv.Conv	[128, 128, 3, 2]
21	[-1, 10] 1	0	ultralytics.nn.modules.conv.Concat	[1]
22	-1 1	378880	ultralytics.nn.modules.block.C3k2	[384, 256, 1, True]
23	[16, 19, 22] 1	684610	ultralytics.nn.modules.head.Segment	[6, 32, 64, 16, None, [64, 128, 256]]

YOLO11n-seg summary: 204 layers, 2,843,778 parameters, 2,843,762 gradients, 9.7 GFLOPs

Transferred 510/561 items from pretrained weights

Freezing layer 'model.23.dfl.conv.weight'

AMP: running Automatic Mixed Precision (AMP) checks...

Downloading <https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo26n.pt> to 'yolo26n.pt': 100%  5.3MB 5.6MB/s 0.9s

AMP: checks passed

train: Fast image access (ping: 0.10.1 ms, read: 200.162.9 MB/s, size: 71.8 KB)

train: Scanning D:\Hydroponics AI\dataset\train\labels... 2870 images, 0 backgrounds, 0 corrupt: 100%  2870/2870 189.8it/s 15.1s

train: D:\Hydroponics AI\dataset\train\images\IMG20230529175058_01_jpg.rf.689ec6c2def314ca4a179b708d4e6646.jpg: 1 duplicate labels removed

train: D:\Hydroponics AI\dataset\train\images\IMG20230529175058_01_jpg.rf.7349d456f89582e02e06c000060c6993.jpg: 1 duplicate labels removed

train: New cache created: D:\Hydroponics AI\dataset\train\labels.cache

WARNING train: 4.9GB RAM required to cache images with 50% safety margin but only 1.6/15.6GB available, not caching images

val: Fast image access (ping: 0.10.0 ms, read: 3.51.1 MB/s, size: 81.9 KB)

val: Scanning D:\Hydroponics AI\dataset\valid\labels... 107 images, 0 backgrounds, 0 corrupt: 100%  107/107 93.5it/s 1.1s

val: New cache created: D:\Hydroponics AI\dataset\valid\labels.cache

WARNING val: 0.2GB RAM required to cache images with 50% safety margin but only 0.0/15.6GB available, not caching images

optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937' and determining best 'optimizer', 'lr0' and 'momentum' automatically...

optimizer: AdamW(lr=0.001, momentum=0.9) with parameter groups 90 weight(decay=0.0), 101 weight(decay=0.0005), 100 bias(decay=0.0)

Plotting labels to D:\Hydroponics AI\runs\segment\runs\baseline_yolo11n_seg\labels.jpg...

C:\Users\HP\AppData\Local\Programs\Python\Python312\Lib\site-packages\torch\jit_trace.py:687: UserWarning: The input to trace is already a ScriptModule, tracing it is a no-op. Returning the object as is.

warnings.warn(

TensorBoard: model graph visualization added

Image sizes 640 train, 640 val

Using 8 dataloader workers

Logging results to D:\Hydroponics AI\runs\segment\runs\baseline_yolo11n_seg

Starting training for 100 epochs...

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df_l_loss	sem_loss	Instances	Size
1/100	4.39G	1.02	2.066	2.7	1.251	0	201	640: 100% ———
————— 180/180 1.5it/s 1:58								
Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	————— 4/4 2.3s/it 9.1s						
all	107	1594	0.566	0.671	0.634	0.494	0.564	0.67
0.636								
0.476								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	df_l_loss	sem_loss	Instances	Size
2/100	3.88G	0.8349	1.473	1.29	1.122	0	152	640: 100% ———
————— 180/180 1.7it/s 1:48								
Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	————— 4/4 1.1it/s 3.6s						

	all	107	1594	0.793	0.777	0.844	0.684	0.796	0.779
0.846	0.639								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
3/100	3.75G	0.7723	1.393	1.005	1.073	0	129	640: 100% —
180/180 1.8it/s 1:41								

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.1it/s 3.5s						

	all	107	1594	0.747	0.794	0.844	0.667	0.764	0.781
0.849	0.604								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
4/100	3.75G	0.7454	1.355	0.9087	1.049	0	125	640: 100% —
180/180 1.8it/s 1:40								

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.2it/s 3.4s						

	all	107	1594	0.665	0.767	0.773	0.646	0.725	0.703
0.775	0.609								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
5/100	3.75G	0.7356	1.315	0.8408	1.039	0	143	640: 100% —
180/180 1.8it/s 1:40								

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.2it/s 3.4s						

	all	107	1594	0.803	0.836	0.888	0.743	0.805	0.838
0.891	0.7								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
6/100	3.82G	0.71	1.293	0.7865	1.027	0	147	640: 100% —
180/180 1.8it/s 1:40								

	Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100% ————— 4/4 1.2it/s 3.3s									
	all	107	1594	0.869	0.855	0.916	0.77	0.867	0.852	0.914
0.725										

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
7/100	3.7G	0.697	1.262	0.7576	1.024	0	134	640: 100% ———
————— 180/180 1.8it/s 1:39								

	Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100% ————— 4/4 1.2it/s 3.4s									
	all	107	1594	0.85	0.854	0.914	0.779	0.849	0.855	0.915
0.739										

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
8/100	3.94G	0.6828	1.24	0.7283	1.017	0	165	640: 100% ———
————— 180/180 1.8it/s 1:40								

	Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100% ————— 4/4 1.2it/s 3.4s									
	all	107	1594	0.816	0.831	0.901	0.763	0.82	0.833	0.902
0.717										

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
9/100	3.77G	0.6709	1.225	0.7065	1.007	0	165	640: 100% ———
————— 180/180 1.8it/s 1:42								

	Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100% ————— 4/4 1.2it/s 3.3s									
	all	107	1594	0.88	0.903	0.947	0.812	0.882	0.905	0.948
0.763										

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
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10/100	3.76G	0.6669	1.215	0.6953	1.006	0	179	640: 100%	——
180/180 1.8it/s 1:40									
Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100%	4/4 1.2it/s 3.3s							
all	107	1594	0.857	0.842	0.924	0.793	0.854	0.839	
0.922	0.746								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size	
11/100	3.78G	0.658	1.194	0.6719	1.001	0	124	640: 100%	——
180/180 1.8it/s 1:40									
Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100%	4/4 1.2it/s 3.3s							
all	107	1594	0.846	0.863	0.92	0.792	0.845	0.862	0.92
0.741									

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size	
12/100	3.86G	0.6396	1.179	0.6567	0.9871	0	141	640: 100%	——
180/180 1.8it/s 1:41									
Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100%	4/4 1.2it/s 3.3s							
all	107	1594	0.874	0.899	0.947	0.819	0.864	0.906	
0.944	0.769								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size	
13/100	3.76G	0.6251	1.15	0.6297	0.9786	0	108	640: 100%	——
180/180 1.7it/s 1:45									
Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100%	4/4 1.2it/s 3.3s							
all	107	1594	0.888	0.892	0.943	0.824	0.881	0.906	
0.946	0.786								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
14/100	3.83G	0.6186	1.147	0.6112	0.9752	0	116	640: 100% —
180/180 1.5it/s 1:57								

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.2it/s 3.3s						
all	107	1594	0.891	0.888	0.949	0.829	0.9	0.879 0.949

0.781

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
15/100	3.86G	0.6097	1.13	0.6069	0.9708	0	122	640: 100% —
180/180 1.5it/s 1:59								

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.2it/s 3.2s						
all	107	1594	0.898	0.904	0.947	0.828	0.897	0.904

0.947 0.779

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
16/100	3.88G	0.6107	1.13	0.5914	0.9712	0	112	640: 100% —
180/180 1.5it/s 2:01								

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.2it/s 3.2s						
all	107	1594	0.899	0.887	0.951	0.828	0.898	0.886 0.95

0.78

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
17/100	3.94G	0.6044	1.124	0.5822	0.968	0	132	640: 100% —
180/180 1.6it/s 1:53								

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.3it/s 3.1s						
all	107	1594	0.88	0.909	0.948	0.835	0.881	0.91 0.948

0.796

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
18/100	3.82G	0.5925	1.1	0.5749	0.9632	0	98	640: 100% —
180/180 1.7it/s 1:43								
Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.3it/s 3.0s						
all	107	1594	0.898	0.895	0.951	0.826	0.893	0.95
0.788								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
19/100	3.85G	0.5914	1.095	0.5645	0.9627	0	119	640: 100% —
180/180 1.8it/s 1:40								
Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.2it/s 3.2s						
all	107	1594	0.884	0.913	0.95	0.837	0.882	0.95
0.794								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
20/100	3.88G	0.5868	1.094	0.5629	0.9614	0	214	640: 100% —
180/180 1.8it/s 1:42								
Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.3it/s 3.0s						
all	107	1594	0.876	0.926	0.955	0.842	0.901	0.9
0.803								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
21/100	3.78G	0.5859	1.1	0.5588	0.9615	0	179	640: 100% —
180/180 1.8it/s 1:42								
Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.2it/s 3.3s						

all 107 1594 0.889 0.909 0.952 0.835 0.891 0.911
0.954 0.79

Epoch GPU_mem box_loss seg_loss cls_loss dfl_loss sem_loss Instances Size
22/100 3.87G 0.5792 1.077 0.5417 0.9567 0 149 640: 100% —
180/180 1.8it/s 1:41

Class Images Instances Box(P R mAP50 mAP50-95) Mask(P R
mAP50 mAP50-95): 100% 4/4 1.3it/s 3.0s

all 107 1594 0.913 0.907 0.956 0.85 0.916 0.908 0.958
0.81

Epoch GPU_mem box_loss seg_loss cls_loss dfl_loss sem_loss Instances Size
23/100 3.73G 0.5786 1.073 0.5442 0.9564 0 118 640: 100% —
180/180 1.8it/s 1:42

Class Images Instances Box(P R mAP50 mAP50-95) Mask(P R
mAP50 mAP50-95): 100% 4/4 1.3it/s 3.1s

all 107 1594 0.91 0.905 0.955 0.845 0.909 0.903 0.954
0.806

Epoch GPU_mem box_loss seg_loss cls_loss dfl_loss sem_loss Instances Size
24/100 3.71G 0.5681 1.057 0.5322 0.9521 0 163 640: 100% —
180/180 1.8it/s 1:41

Class Images Instances Box(P R mAP50 mAP50-95) Mask(P R
mAP50 mAP50-95): 100% 4/4 1.3it/s 3.0s

all 107 1594 0.899 0.917 0.959 0.857 0.9 0.917 0.959
0.809

Epoch GPU_mem box_loss seg_loss cls_loss dfl_loss sem_loss Instances Size
25/100 3.79G 0.5663 1.055 0.5214 0.9493 0 200 640: 100% —
180/180 1.8it/s 1:42

	Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100%				4/4 1.3it/s 3.0s					
	all	107	1594	0.898	0.918	0.958	0.856	0.899	0.917	
	0.959	0.817								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
26/100	3.89G	0.5606	1.043	0.5176	0.949	0	214	640: 100%
180/180 1.8it/s 1:42								

	Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100%				4/4 1.3it/s 3.1s					
	all	107	1594	0.926	0.909	0.954	0.855	0.919	0.919	
	0.957	0.816								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
27/100	4.02G	0.5603	1.037	0.5136	0.9491	0	117	640: 100%
180/180 1.8it/s 1:42								

	Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100%				4/4 1.3it/s 3.0s					
	all	107	1594	0.917	0.9	0.952	0.856	0.918	0.901	
	0.815	0.954								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
28/100	3.78G	0.553	1.033	0.5069	0.9446	0	139	640: 100%
180/180 1.8it/s 1:41								

	Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100%				4/4 1.3it/s 3.0s					
	all	107	1594	0.92	0.906	0.955	0.856	0.92	0.908	
	0.814	0.955								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
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29/100	3.71G	0.5517	1.026	0.5043	0.945	0	119	640: 100%	——
180/180 1.8it/s 1:42									
Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100%	4/4 1.4it/s 3.0s							
all	107	1594	0.909	0.914	0.958	0.857	0.913	0.911	
0.959	0.814								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size	
30/100	3.91G	0.5487	1.025	0.5039	0.944	0	120	640: 100%	——
180/180 1.8it/s 1:40									
Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100%	4/4 1.3it/s 3.1s							
all	107	1594	0.886	0.909	0.95	0.852	0.887	0.909	0.95
0.793									

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size	
31/100	3.84G	0.5472	1.023	0.501	0.9429	0	99	640: 100%	——
180/180 1.8it/s 1:41									
Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100%	4/4 1.3it/s 3.0s							
all	107	1594	0.915	0.915	0.955	0.858	0.916	0.915	
0.954	0.819								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size	
32/100	3.72G	0.5406	1.013	0.4954	0.9396	0	102	640: 100%	——
180/180 1.7it/s 1:44									
Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100%	4/4 1.2it/s 3.2s							
all	107	1594	0.913	0.929	0.96	0.865	0.914	0.93	0.962
0.824									

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
33/100	3.78G	0.5379	1.001	0.4845	0.9373	0	165	640: 100% —
180/180 1.8it/s 1:42								

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.3it/s 3.0s						
all	107	1594	0.912	0.93	0.964	0.874	0.912	0.931
0.833								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
34/100	3.71G	0.5341	0.9991	0.4867	0.9395	0	152	640: 100% —
180/180 1.8it/s 1:42								

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.3it/s 3.0s						
all	107	1594	0.921	0.917	0.961	0.87	0.923	0.92
0.828								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
35/100	3.88G	0.5303	0.9844	0.4768	0.936	0	138	640: 100% —
180/180 1.8it/s 1:42								

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.3it/s 3.0s						
all	107	1594	0.911	0.922	0.958	0.876	0.914	0.925
0.837								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
36/100	3.68G	0.5335	0.9949	0.4802	0.9364	0	223	640: 100% —
180/180 1.8it/s 1:42								

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.4it/s 2.9s						
all	107	1594	0.895	0.927	0.959	0.874	0.896	0.928
0.959 0.836								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
37/100	3.96G	0.5286	0.9793	0.4747	0.9338	0	138	640: 100% —
180/180 1.8it/s 1:42								

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.3it/s 3.0s						
all	107	1594	0.925	0.931	0.964	0.883	0.927	0.933
0.965	0.835							

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
38/100	3.74G	0.525	0.9801	0.47	0.9327	0	169	640: 100% —
180/180 1.8it/s 1:41								

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.3it/s 3.0s						
all	107	1594	0.927	0.913	0.961	0.874	0.928	0.914
0.963	0.834							

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
39/100	3.84G	0.526	0.9788	0.4686	0.9307	0	140	640: 100% —
180/180 1.8it/s 1:41								

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.3it/s 3.0s						
all	107	1594	0.909	0.921	0.956	0.874	0.928	0.909
0.959	0.836							

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
40/100	3.71G	0.5191	0.9672	0.4621	0.927	0	112	640: 100% —
180/180 1.8it/s 1:42								

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.3it/s 3.0s						

all 107 1594 0.91 0.941 0.96 0.873 0.911 0.943 0.961
0.833

Epoch GPU_mem box_loss seg_loss cls_loss dfl_loss sem_loss Instances Size
41/100 3.82G 0.5178 0.9692 0.4622 0.9288 0 182 640: 100% —
180/180 1.8it/s 1:41

Class Images Instances Box(P R mAP50 mAP50-95) Mask(P R
mAP50 mAP50-95): 100% 4/4 1.2it/s 3.2s

all 107 1594 0.925 0.93 0.964 0.88 0.919 0.939 0.966
0.843

Epoch GPU_mem box_loss seg_loss cls_loss dfl_loss sem_loss Instances Size
42/100 3.82G 0.5163 0.9714 0.4597 0.9272 0 175 640: 100% —
180/180 1.8it/s 1:42

Class Images Instances Box(P R mAP50 mAP50-95) Mask(P R
mAP50 mAP50-95): 100% 4/4 1.3it/s 3.0s

all 107 1594 0.915 0.933 0.963 0.879 0.916 0.933
0.963 0.838

Epoch GPU_mem box_loss seg_loss cls_loss dfl_loss sem_loss Instances Size
43/100 3.73G 0.5073 0.9488 0.4508 0.9238 0 142 640: 100% —
180/180 1.8it/s 1:42

Class Images Instances Box(P R mAP50 mAP50-95) Mask(P R
mAP50 mAP50-95): 100% 4/4 1.3it/s 3.0s

all 107 1594 0.93 0.924 0.962 0.883 0.93 0.924 0.962
0.846

Epoch GPU_mem box_loss seg_loss cls_loss dfl_loss sem_loss Instances Size
44/100 3.98G 0.515 0.9585 0.4521 0.9251 0 167 640: 100% —
180/180 1.8it/s 1:41

	Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100%					4/4 1.4it/s 3.0s				
	all	107	1594	0.919	0.943	0.966	0.89	0.919	0.943	0.966
0.852										

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
45/100	3.87G	0.5078	0.9568	0.4484	0.9232	0	154	640: 100% —
180/180 1.8it/s 1:42								

	Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100%					4/4 1.4it/s 2.9s				
	all	107	1594	0.925	0.927	0.964	0.884	0.928	0.929	
0.967 0.846										

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
46/100	3.85G	0.5009	0.9425	0.443	0.921	0	105	640: 100% —
180/180 1.5it/s 1:59								

	Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100%					4/4 1.4s/it 5.7s				
	all	107	1594	0.908	0.934	0.963	0.883	0.91	0.935	0.964
0.839										

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
47/100	3.89G	0.4999	0.9359	0.4435	0.9222	0	163	640: 100% —
180/180 1.0s/it 3:01								

	Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100%					4/4 1.3it/s 3.1s				
	all	107	1594	0.918	0.937	0.965	0.886	0.919	0.937	
0.966 0.847										

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
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48/100	3.83G	0.4972	0.9326	0.4409	0.9179	0	118	640: 100%	—
180/180 1.0it/s 2:52									
Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100%	4/4 1.5s/it 5.9s							
all	107	1594	0.926	0.945	0.966	0.887	0.927	0.945	
0.967	0.851								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size	
49/100	3.77G	0.4991	0.9377	0.44	0.9184	0	232	640: 100%	—
180/180 1.1s/it 3:22									
Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100%	4/4 1.4s/it 5.5s							
all	107	1594	0.925	0.926	0.963	0.886	0.926	0.927	
0.965	0.843								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size	
50/100	3.85G	0.5	0.9418	0.436	0.9206	0	98	640: 100%	—
180/180 1.1s/it 3:15									
Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100%	4/4 1.3s/it 5.3s							
all	107	1594	0.908	0.934	0.964	0.886	0.908	0.934	
0.965	0.844								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size	
51/100	3.7G	0.4945	0.9321	0.4352	0.919	0	130	640: 100%	—
180/180 1.1s/it 3:16									
Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100%	4/4 1.3s/it 5.4s							
all	107	1594	0.917	0.943	0.966	0.888	0.918	0.944	
0.967	0.848								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
52/100	3.86G	0.4906	0.9236	0.4284	0.916	0	193	640: 100% —
180/180 1.3s/it 3:46								

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.8s/it 7.3s						
all	107	1594	0.923	0.933	0.962	0.886	0.924	0.933
0.964	0.846							

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
53/100	3.91G	0.4884	0.915	0.4241	0.9142	0	128	640: 100% —
180/180 1.4s/it 4:07								

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.9s/it 7.6s						
all	107	1594	0.929	0.93	0.964	0.886	0.93	0.931
0.848	0.965							

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
54/100	3.81G	0.4887	0.9276	0.4262	0.9147	0	135	640: 100% —
180/180 1.4s/it 4:12								

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.8s/it 7.4s						
all	107	1594	0.905	0.936	0.958	0.886	0.907	0.938
0.846	0.96							

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
55/100	3.76G	0.4898	0.9199	0.4205	0.9144	0	148	640: 100% —
180/180 1.4s/it 4:10								

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 2.0s/it 8.0s						
all	107	1594	0.931	0.939	0.964	0.892	0.932	0.94
0.852	0.966							

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
56/100	3.7G	0.4852	0.8988	0.4184	0.9146	0	131	640: 100% —
180/180 1.2s/it 3:42								
Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.4s/it 5.6s						
all	107	1594	0.926	0.941	0.966	0.893	0.93	0.943
0.86								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
57/100	3.71G	0.4881	0.9146	0.4204	0.9138	0	148	640: 100% —
180/180 1.1it/s 2:44								
Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.2it/s 3.2s						
all	107	1594	0.935	0.942	0.967	0.896	0.935	0.943
0.968 0.854								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
58/100	3.79G	0.477	0.8983	0.4118	0.9094	0	155	640: 100% —
180/180 1.6it/s 1:54								
Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.1it/s 3.5s						
all	107	1594	0.927	0.937	0.966	0.896	0.929	0.939
0.968 0.858								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
59/100	3.9G	0.4804	0.9022	0.4111	0.9096	0	144	640: 100% —
180/180 1.6it/s 1:51								
Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.1it/s 3.5s						

	all	107	1594	0.919	0.953	0.966	0.897	0.921	0.955
0.967	0.856								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
60/100	3.79G	0.4767	0.8914	0.4085	0.9068	0	132	640: 100% —
180/180 1.5it/s 1:59								

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100% — 4/4 1.3it/s 3.1s							

	all	107	1594	0.931	0.939	0.967	0.899	0.931	0.939
0.967	0.855								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
61/100	3.79G	0.4715	0.8872	0.4029	0.9053	0	115	640: 100% —
180/180 1.7it/s 1:47								

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100% — 4/4 1.3it/s 3.0s							

	all	107	1594	0.935	0.929	0.964	0.894	0.936	0.93	0.963
0.851										

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
62/100	3.75G	0.4739	0.891	0.4064	0.9079	0	170	640: 100% —
180/180 1.5it/s 1:58								

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100% — 4/4 1.3it/s 3.2s							

	all	107	1594	0.937	0.932	0.967	0.9	0.938	0.933	0.968
0.856										

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
63/100	3.77G	0.4737	0.8823	0.4041	0.9059	0	140	640: 100% —
180/180 1.8it/s 1:41								

	Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100% ————— 4/4 1.3it/s 3.1s								
	all	107	1594	0.937	0.935	0.966	0.898	0.938	0.936
						0.966			0.854

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
64/100	3.85G	0.4663	0.8795	0.401	0.9036	0	145	640: 100% —
————— 180/180 1.8it/s 1:42								

	Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100% ————— 4/4 1.3it/s 3.0s								
	all	107	1594	0.925	0.949	0.968	0.898	0.925	0.95
									0.969
									0.855

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
65/100	3.74G	0.4695	0.8797	0.3998	0.9065	0	122	640: 100% —
————— 180/180 1.8it/s 1:41								

	Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100% ————— 4/4 1.3it/s 3.0s								
	all	107	1594	0.921	0.945	0.964	0.897	0.924	0.948
									0.967
									0.859

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
66/100	3.79G	0.4668	0.8781	0.3976	0.9037	0	145	640: 100% —
————— 180/180 1.8it/s 1:42								

	Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100% ————— 4/4 1.3it/s 3.0s								
	all	107	1594	0.922	0.942	0.968	0.898	0.924	0.942
									0.968
									0.86

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
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67/100 3.82G 0.4698 0.8807 0.4001 0.9027 0 147 640: 100% —
180/180 1.8it/s 1:42

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%				4/4	1.3it/s 3.1s		

all	107	1594	0.926	0.935	0.965	0.897	0.927	0.936
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0.966 0.861

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
68/100	3.75G	0.4681	0.877	0.3954	0.9047	0	109	640: 100% —

180/180 1.8it/s 1:41

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%				4/4	1.3it/s 3.0s		

all	107	1594	0.934	0.927	0.965	0.9	0.936	0.928 0.965
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0.858

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
69/100	3.83G	0.4606	0.8626	0.3951	0.9014	0	141	640: 100% —

180/180 1.8it/s 1:42

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%				4/4	1.3it/s 3.2s		

all	107	1594	0.929	0.942	0.968	0.899	0.93	0.943 0.969
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0.861

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
70/100	3.83G	0.462	0.8691	0.3958	0.9038	0	137	640: 100% —

180/180 1.8it/s 1:41

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%				4/4	1.4it/s 2.9s		

all	107	1594	0.918	0.93	0.962	0.898	0.919	0.931 0.963
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0.859

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
71/100	3.77G	0.455	0.8601	0.3857	0.8993	0	146	640: 100% —
180/180 1.8it/s 1:41								
Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.2it/s 3.2s						
all	107	1594	0.926	0.936	0.967	0.903	0.928	0.938
0.968	0.865							

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
72/100	3.75G	0.4559	0.862	0.3846	0.897	0	131	640: 100% —
180/180 1.8it/s 1:41								
Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.3it/s 3.0s						
all	107	1594	0.93	0.944	0.965	0.901	0.938	0.939
0.864								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
73/100	3.88G	0.452	0.8513	0.3848	0.8979	0	158	640: 100% —
180/180 1.8it/s 1:41								
Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.3it/s 3.0s						
all	107	1594	0.932	0.928	0.965	0.901	0.932	0.929
0.967	0.865							

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
74/100	3.82G	0.4529	0.8567	0.3829	0.8996	0	121	640: 100% —
180/180 1.8it/s 1:41								
Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.3it/s 3.1s						
all	107	1594	0.936	0.941	0.966	0.902	0.94	0.941
0.86								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
75/100	3.78G	0.4545	0.8562	0.3806	0.8969	0	157	640: 100% —
180/180 1.7it/s 1:43								

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.3it/s 3.0s						
all	107	1594	0.936	0.942	0.968	0.902	0.936	0.943
0.968	0.864							

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
76/100	3.88G	0.4501	0.8559	0.3794	0.8969	0	144	640: 100% —
180/180 1.8it/s 1:41								

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.3it/s 3.0s						
all	107	1594	0.932	0.939	0.968	0.905	0.933	0.94
0.866	0.968							

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
77/100	3.86G	0.4475	0.849	0.3761	0.8958	0	170	640: 100% —
180/180 1.8it/s 1:42								

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.3it/s 3.0s						
all	107	1594	0.935	0.94	0.969	0.906	0.936	0.94
0.867	0.969							

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
78/100	3.87G	0.4474	0.8447	0.373	0.8977	0	201	640: 100% —
180/180 1.8it/s 1:41								

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.3it/s 3.1s						

	all	107	1594	0.932	0.944	0.965	0.903	0.933	0.945
0.967	0.862								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
79/100	3.86G	0.4467	0.8393	0.3725	0.8946	0	120	640: 100% —
180/180 1.8it/s 1:41								

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.4it/s 2.9s						

	all	107	1594	0.925	0.943	0.967	0.904	0.926	0.945
0.968	0.867								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
80/100	3.86G	0.4415	0.8423	0.3712	0.8937	0	178	640: 100% —
180/180 1.8it/s 1:41								

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.3it/s 3.0s						

	all	107	1594	0.936	0.929	0.964	0.904	0.937	0.93	0.966
0.863										

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
81/100	3.9G	0.442	0.8344	0.3697	0.8938	0	116	640: 100% —
180/180 1.8it/s 1:41								

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.4it/s 3.0s						

	all	107	1594	0.94	0.938	0.966	0.903	0.941	0.939	0.967
0.865										

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
82/100	3.77G	0.4375	0.8267	0.3668	0.8926	0	141	640: 100% —
180/180 1.8it/s 1:40								

	Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100%				4/4 1.3it/s 3.0s					
	all	107	1594	0.938	0.941	0.967	0.906	0.935	0.944	
	0.967	0.863								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
83/100	3.83G	0.4389	0.8306	0.367	0.892	0	90	640: 100% —
180/180 1.8it/s 1:41								

	Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100%				4/4 1.4it/s 2.9s					
	all	107	1594	0.947	0.94	0.967	0.907	0.947	0.94	
	0.867	0.968								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
84/100	3.79G	0.4373	0.8321	0.3656	0.8913	0	141	640: 100% —
180/180 1.8it/s 1:41								

	Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100%				4/4 1.4it/s 2.9s					
	all	107	1594	0.931	0.942	0.966	0.908	0.932	0.943	
	0.967	0.868								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
85/100	3.82G	0.4367	0.8304	0.3613	0.8896	0	121	640: 100% —
180/180 1.8it/s 1:41								

	Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100%				4/4 1.3it/s 3.0s					
	all	107	1594	0.938	0.945	0.969	0.91	0.939	0.945	
	0.87	0.97								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
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86/100	3.79G	0.4385	0.8309	0.3624	0.8914	0	94	640: 100%	——
180/180 1.8it/s 1:43									
Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100%	4/4 1.3it/s 3.2s							
all	107	1594	0.933	0.944	0.965	0.907	0.935	0.946	
0.967	0.867								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size	
87/100	3.84G	0.4319	0.8231	0.3615	0.8892	0	125	640: 100%	—
180/180 1.8it/s 1:41									
Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100%	4/4 1.3it/s 3.0s							
all	107	1594	0.934	0.939	0.966	0.908	0.936	0.94	0.967
0.87									

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size	
88/100	3.84G	0.4333	0.8241	0.3644	0.8903	0	152	640: 100%	—
180/180 1.8it/s 1:41									
Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100%	4/4 1.3it/s 3.1s							
all	107	1594	0.936	0.936	0.967	0.91	0.937	0.937	0.967
0.871									

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size	
89/100	3.79G	0.4325	0.8173	0.356	0.8882	0	138	640: 100%	——
180/180 1.7it/s 1:44									
Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100%	4/4 1.3it/s 3.1s							
all	107	1594	0.93	0.943	0.966	0.907	0.931	0.944	0.967
0.869									

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
90/100	3.84G	0.4326	0.8219	0.3568	0.8889	0	166	640: 100% —

180/180 1.7it/s 1:44

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.3it/s 3.2s						
all	107	1594	0.934	0.944	0.966	0.909	0.935	0.946
0.968	0.87							

Closing dataloader mosaic

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
91/100	3.2G	0.3637	0.7318	0.2852	0.8596	0	58	640: 100% —

180/180 1.9it/s 1:32

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.1it/s 3.5s						
all	107	1594	0.934	0.942	0.966	0.903	0.933	0.945
0.968	0.863							

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
92/100	3.19G	0.3477	0.7006	0.2682	0.8494	0	89	640: 100% —

180/180 2.1it/s 1:27

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.2it/s 3.3s						
all	107	1594	0.939	0.945	0.969	0.91	0.94	0.945
0.869								

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
93/100	3.31G	0.3455	0.696	0.2653	0.8526	0	86	640: 100% —

180/180 2.0it/s 1:29

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%	4/4 1.3it/s 3.2s						

all 107 1594 0.94 0.945 0.968 0.911 0.94 0.945 0.969
0.87

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
94/100	3.17G	0.3444	0.6897	0.263	0.85	0	86	640: 100% —
180/180 1.9it/s 1:35								

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100% — 4/4 1.3it/s 3.1s							

all 107 1594 0.939 0.942 0.968 0.91 0.94 0.943 0.969
0.87

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
95/100	3.17G	0.3391	0.6833	0.2578	0.8467	0	71	640: 100% —
180/180 1.9it/s 1:35								

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100% — 4/4 1.2it/s 3.4s							

all 107 1594 0.94 0.945 0.969 0.912 0.941 0.946 0.97
0.872

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
96/100	3.24G	0.3401	0.6838	0.258	0.8488	0	88	640: 100% —
180/180 1.9it/s 1:33								

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100% — 4/4 1.2it/s 3.3s							

all 107 1594 0.941 0.946 0.968 0.913 0.942 0.947
0.969 0.873

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
97/100	3.12G	0.3339	0.677	0.2539	0.8469	0	72	640: 100% —
180/180 1.9it/s 1:33								

	Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100% ————— 4/4 1.2it/s 3.4s									
	all	107	1594	0.944	0.947	0.97	0.914	0.945	0.948	0.971

0.874

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
98/100	3.14G	0.3359	0.6771	0.2563	0.8477	0	93	640: 100% ———

————— 180/180 2.0it/s 1:30

	Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100% ————— 4/4 1.2it/s 3.3s									
	all	107	1594	0.936	0.947	0.969	0.912	0.937	0.948	0.97

0.874

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
99/100	3.17G	0.3303	0.6697	0.251	0.8451	0	84	640: 100% ———

————— 180/180 1.9it/s 1:34

	Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100% ————— 4/4 1.1it/s 3.6s									
	all	107	1594	0.945	0.94	0.969	0.915	0.947	0.941	0.971

0.875

Epoch	GPU_mem	box_loss	seg_loss	cls_loss	dfl_loss	sem_loss	Instances	Size
100/100	3.22G	0.3336	0.6744	0.2509	0.844	0	77	640: 100% ———

————— 180/180 1.8it/s 1:39

	Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R	
mAP50	mAP50-95): 100% ————— 4/4 1.1it/s 3.8s									
	all	107	1594	0.936	0.947	0.969	0.914	0.937	0.948	0.97

0.874

100 epochs completed in 3.363 hours.

Optimizer stripped from D:\Hydroponics

AI\runs\segment\runs\baseline_yolo11n_seg\weights\last.pt, 6.0MB

Optimizer stripped from D:\Hydroponics

AI\runs\segment\runs\baseline_yolo11n_seg\weights\best.pt, 6.0MB

Validating D:\Hydroponics AI\runs\segment\runs\baseline_yolo11n_seg\weights\best.pt...

Ultralytics 8.4.35 Python-3.12.5 torch-2.5.1+cu121 CUDA:0 (NVIDIA GeForce RTX 3050 6GB Laptop GPU, 6144MiB)

YOLO11n-seg summary (fused): 114 layers, 2,835,738 parameters, 0 gradients, 9.6 GFLOPs

	Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	Mask(P	R
mAP50	mAP50-95): 100%					4/4	2.2s/it	8.6s	
	all	107	1594	0.942	0.943	0.969	0.915	0.946	0.941
0.875									
	Calcium Deficiency		43	294	0.915	0.956	0.963	0.91	0.921
0.968	0.884								0.959
	Healthy	58	311	0.909	0.797	0.91	0.795	0.921	0.791
0.913	0.768								
	Magnesium Deficiency		25	198	0.991	0.965	0.988	0.945	0.994
0.965	0.988	0.873							
	Nitrogen Deficiency		30	258	0.93	0.984	0.989	0.97	0.93
0.989	0.941								0.979
	Phosphorus Deficiency		38	291	0.94	0.973	0.981	0.91	0.94
0.981	0.875								0.968
	Potassium Deficiency		44	242	0.968	0.983	0.984	0.958	0.97
0.984	0.908								0.983

Speed: 0.8ms preprocess, 9.4ms inference, 0.0ms loss, 14.2ms postprocess per image

Results saved to D:\Hydroponics AI\runs\segment\runs\baseline_yolo11n_seg

Learn more at <https://docs.ultralytics.com/modes/train>

PS D:\Hydroponics AI>